EFFECTIVE PHARMACEUTICAL DRUG TREATMENT AGAINST SARS-CoV-2 INFECTION

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Abstract:

Corona virus disease 2019 (Covid-19) is one of the most severe pandemics declared as Public Health emergency of international concern by World Health Organization on 20th January 2020, responsible for millions of deaths and still peoples are suffering symptomatically and asymptomatically along with number of secondary infections. As priority in the therapeutic management various structures of drugs, their mechanism of action (Inhibition of viral RNA Polymerase enzyme, endocytic pathway, sialic acid receptor mechanism and antiviral effect) in relation with the location of virus and their destruction, applications and side effects were studied and found Remdesivir, Chloroquine, Hydroxychloroquine and Favipiravirare effective drugs for the treatment of SARS CoV-2 virus. Also, convalescent plasma therapy which is classic adaptive immuno-therapy largely applied for the prevention and treatment of Covid-19.

Keywords: SARS CoV-2, ACE2, Remdesivir, Plasma therapy, S Protein, Immunomodulators

Introduction:

From the last manycenturies deadliest pandemic had taken place which causes death of the many people in the world. Some of them are Justinian plague (541-542 B.C.) caused due to *Yersinia pestis*-nearly about 40 million deaths i.e. 19.1% of the population had loss their life in this pandemic. Highly contagious Smallpoxdisease caused due to Variola virus in which skin eruption like pustules form and mortality rate was 30% and Edward Jenner had developed the world first vaccine in the 1798. In the first decade of last century in the year 1918-1920, Spanish flu occurred in the setting of modern medicine and at this time the H1N1 strain of the Influenza virus had spread to every corner of the world around 45 million of individual lost their life. After this Spanish flu - SARS (Severe Acute Respiratory Syndrome) was the first outbreak in the Twenty-first century in the year 2002-2003 and approximately 770 people loss their life in this outbreak. Then, first case of Ebola virus (2013) found in the central and west Africa and

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outbreak in the remote in Guniea in the December 2013. There were about 28,000 cases and the 11,000 fatalities. After that Zika virus, a little dormant virus found in the rhesus monkey in the Uganda in 2014 only outbreak among human recorded in Micronesia in 2007 and virus was identified in the Brazil in 2015 (Damir, 2019).

Currently, Corona virus disease 2019 (Covid-19) outbreak was in Wuhan City, Hubei Province, China in the month of December 2020, caused by a novel severe acute respiratory syndrome corona virus 2 (SARS-CoV-2) and on 30th January 2020, World Health Organization (WHO) declared the outbreak as Public Health Emergency of International Concern (Harapan et. al Covid-19 virus is highly infectious and caused high mortality across the world led to 3.5 million deaths and may more death in near future, if proper therapeutic treatment not given. There are many variants of concern and Variant of interest for giving correct treatment for covid-19 disease. As on 8th June 2021, there have been 173,271,769 confirmed cases of COVID-19, including 3,733,980 deaths, reported to WHO and total of 1,900,955,505 vaccine doses have been administered till 5th June 2021 (https://covid19.who.int/).

Coronaviruses are large, enveloped, positive stranded RNA viruses responsible for infecting a wide variety of mammalian and avian species. If you see the phylogenetic relationship in the Coronavirinae subfamily which is formed by mainly four genera as Alphacoronavirus, Betacoronavirus (four lineages A, B, C and D), Gammacoronavirus and Deltacoronavirus. Corona viruses are non-segmented enveloped virus with single stranded RNA ranging about 26-32 Kb in length. The spherical shape of the corona virus and the size of the virus is about 60-140 nm and the outer surface of the studded with the distinctive 9-10 nm long (Anushmali et. al, 2020). The novel corona virus has four structural proteins. One of them called the spike proteins, mediates the virus entry into a host cell by binding to an enzyme attached to the cell called angiotensin converting enzyme2 (ACE2) (Squarene *et al.,* 2020).Once the virus is infected to the body it attacks the respiratory system then the virus releases its RNA and uses the RdRp complex to replicate its genome.

Symptoms of Covid-19:

The people with Covid-19 had wide range of symptoms from mild symptoms to severe illness, may appear 2-14 days after exposure to the Virus. The symptoms include Fever or chills, Cough, Shortness of breath or difficulty breathing, Fatigue, Muscle or body aches, Headache, New loss of taste or smell, Sore throat, Congestion or runny nose, Nausea or vomiting, Diarrhea, Conjunctivitis, Loss of taste or smell, Rash on skin, or discoloration of fingers or toes (www.cdc.gov/).

Drug Treatment for Covid-19 disease:

The current therapeutic management for covid-19 mainly includes antiviral drugstherapy (Remdesivir, Hydro-chloroquine, Ivermectin and Tocilizumab) Plasma therapy, Immunomodulator agent (corticosteroids, Interferon β-la, Interleukin(IL-1) and Anti-IL-6 receptor monoclonal Antibodies)are used for treatment of covid-19 patients (Cascella *et al.,* 2020). SARS-CoV-2, is a single-stranded RNA-enveloped virus, targets cells through the viral structural spike (S) protein that binds to the angiotensin-converting enzyme 2 (ACE2) receptor. After receptor binding to enter inside the cell, virus particle uses receptors of host cell and endosomes to enter inside cells. A host type 2 transmembrane serine protease, TMPRSS2, provide facility to enter intocell via the S protein (Hoffman *et al.,* 2020). After that synthesis of viral polyproteins takes place which encode for replicase-transcriptase complex. Then virus synthesizes RNA using its RNA-dependent RNA polymerase enzyme.

Structural proteins are synthesized leading to completion of assembly and release of viral particles (Chen *et al.,* 2020; Fehr and Perlman 2015; Fung and Liu, 2014). These viral lifecycle steps provide potential targets for drug therapy. Promising drug targets include nonstructural proteins (e. g, 3-chymotrypsin-like protease, papainlike protease, RNA-dependent RNA polymerase), who share similarity with other novel coronaviruses (nCoVs). The mechanism of action of drugs and major pharmacologic parameters of treatments or adjunctive therapies for COVID-19 is given below (James *et al.,* 2020).

Remdesivir:

Remdesivir is the prolong drug which intracellularly metabolizesand structurally analogue to adenosine triphosphate which inhibit the viral RNA polymerase. The main structural feature distinguishes remdesivir form the adenosine is the modification of specific chemical molecule. This drug has the broad-spectrum activity against Filovirus and Corona virus family members. In-vitro testing observed that remdesivir has the activity against the corona virus (Saqrane *et al.,* 2020).

Structure of Remdesivir:

It is essentially a modified version of the natural component of adenosine which is essential for formation of DNA and RNA. The active form of the drug has the three-phosphate group recognized by the viral RNA polymerase enzyme. When the remdesivir come in the contact with the virus and the virus mistakenly incorporates it into the copies of the RNA instead of the adenosine triphosphate. The compound also evades the proofreading enzyme by slotting itself into the RNA at the slightly different position from where the adenosine triphosphate is supposed to be. Once Remdesivir is incorporated into the RNA growth chain, the presence of

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carbon and nitrogen group can cause the sugar to bend which in turn distorts the shape of the RNA chain, therefore only three additional nucleotides can be added. By adding these nucleotides, it stops the synthesis of RNA strand and eventually interrupts the virus replication mechanism. As the corona virus has specialized mechanism which recognizes the artificial nucleotides and eliminate the artificial nucleoside and prevent the replication of the corona virus (Eastman *et al.,* 2020). Ultimately the RNA synthesis process has be stop. Recently, the U.S. national institute of Allergy and infection disease conducted clinical trials with Remdesivir and found it is effective for the covid-19 virus. U.S. Food and Drug administration granted emergency approval for the use of the remdesivir as an effective antiviral drug against Covid-19 (Clavin *et al.,* 2020).

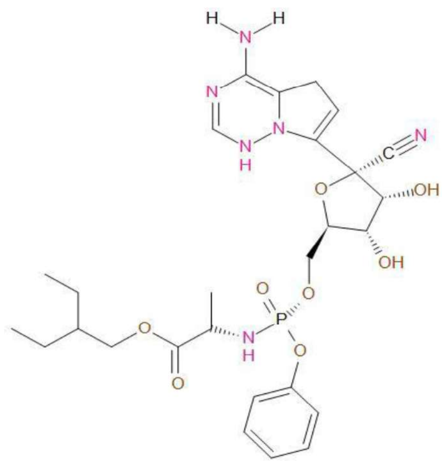

Figure 1: Structure of Remdesivir

Application of the Remdesivir:

It is anintravenous nucleotide drug which contain Adenosine analog which bind to the viral RNA dependent RNA polymerase and inhibit the viral replication through premature termination of RNA activity against SARS-CoV-2 virus.

Side effects of the Remdesivir drug:

1. An infusion reaction may occur with remdesivir with symptoms such as the low blood pressure, nausea, vomiting, sweating and shivering.
2. In the case of the serious allergic reaction symptoms such as the rashes, itching, swelling, severe dizziness.

Chloroquine (CQ) and Hydroxy-Chloroquine (HCQ):

Hydroxy-chloroquine is the safer analogue of the chloroquine which is used as antimalarial and immuno-modulator against the covid-19. It hasantiviral activity in which decrease in the phago-lysosome fusion, increasing intracellular pH and imparting viral receptor glycosylation. Immune-modulatory effect responsible for decreasing production of cytokines especially IL-1 and IL-6 and inhibiting the toll-like receptor (Arshad *et al.,* 2020).

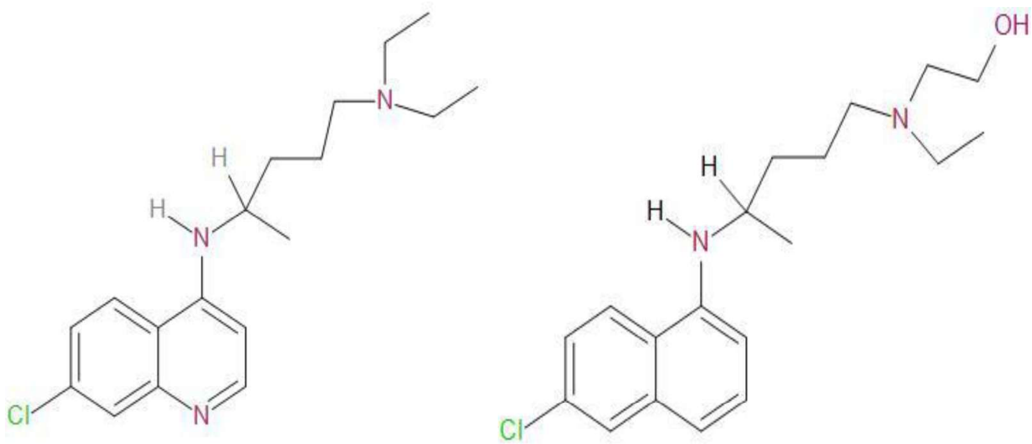

Figure 2: Structure of chloroquine andhydroxy-cholorquine:

Some of the researcher proved that hydroxy derivative is safer than the CQdue to the decrease in the renal and ocular toxicity,so it is used as a substitute for CQ. There are some following ways thatHQC and QC showed some pathway to inhibit the covid -19 virus.

Mechanism of action of CQ and HCQ by the endocytic pathway:

As Covid-19 virus enter into the host cell by endocytic pathways. Pneumocytes are CtBP-1 & 2 and pak proteins are important for the micropinocytosis which is an essential mediator in the Pneumocytes endocytic pathway. As chloroquine get collected in the endosomes and lysosomes,it leads to the pH neutralization and little hindrance in the effect of the proteases, which preventing the S protein cleavage (Satarkar *et al.,* 2020). Chloroquine inhibits the fusion of the lysosomes due todysregulation of Sytaxin which hamper the golgi functioning and block transportation essential material into lysosomes.

Hydroxy-cholorquine prevents the movement of SARS-CoV-2 virus, form the early endosomes to early lysosomes that are important for the viral release in the host. Increases in the pH of the lysosomes and endosomes are mediated by hydroxyl-cholorquine further lead to autophagosomes which break the S protein which prevents the membrane fusion.

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Sialic acid receptorinhibitory mechanism:

Recently, it has been found that MERS-CoV virus binding occur at the receptor of the sialic acidand it identified at the N-terminal of the Sprotein in the SARS-CoV-2. This mediates the entry of the sialic acid receptor in the upper respiratory pathway and this receptor was previously known as ACE-2 receptor. There is another binding site for binding of gangliosides in the SARS-CoV-2, S proteins N- terminal domain (NTD). Especially, 9-O-SIA variant inhibiting the sialic acids by the drug Hydroxy-chloroquine and chloro-quine. However, study proved that hydroxy-chloroquineshows better potency than chloroquine.Epithelial cells of the conjunctiva and cornea has the 2-6 linkage sialic acid receptors as the SARS-CoV-2 is also speculated to enterthrough the binding with alpha 2, 6 linkage and alpha 2, 3 linkages of the sialic acid receptors. Nasolacrimal regionpossess both the receptor therefore respiratory passage via duct of nasolacrimal lead to successful entry into the host cells. (Satarkar *et al.,* 2020).

Application of Hydroxy-chloroquine:-

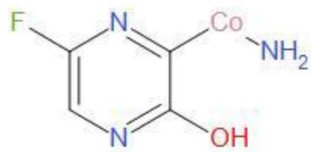
1. This drug decreases pH of the lysosomes and endosomes.
2. used in a treatment or prevention of Malaria.
3. Also employed in the treatment of lupus.
4. It is used in arthritis to stop swelling and pain.

Side effects of Chloroquine and Hydroxy-chloroquine:

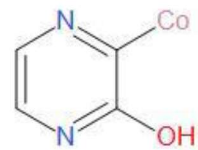
1. A higher dose of the hydroxyl-chloroquine cause the irreversible retina damage which may cause the risk of retinal damage higher in people with the pre-existing eye problem, kidney disease.
2. There are many symptoms such as vision change or damage to your retina, heart disease, heart rhythm disorder, diabetes, a stomach disorder, an allergy to quine, liver or kidney disease, porphyria (a genetic enzyme disorder that cause symptoms affecting the skin or nervous system), headache and Nausea (Alanagreh *et al.,* 2020).

Favipiravir:

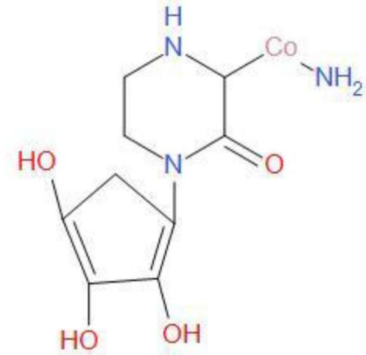
The chemical name of the Favipiravir is 6-Fluoro-3-oxo-3,4-dihydropyrazine-2-carboxamide alternatively, it is also called as T-705, Fapilavir and Favilavir. The class of this antiviral agent. Flavipiravir is derived by chemical modification of the pyrazine moiety of T-1105.



Favipirain (T-705)



T-1105



T-1106

Favipiravir drug is administered as a prodrug which has low distribution of volume and 54% of the protein binding. This drug has a short life cycle about 2.5 to 5 hours leading the rapid renal elimination in hydroxylated form. This elimination is maintained by the enzyme aldehyde oxidase and marginally by Xanthine oxidase. This drug is exhibit in the two form that it is dose dependent and time dependent pharmacokinetics. This drug is not metabolized by the P-450 system but inhibit the component i.e CYP2C8 hence, it is necessary to maintain the precaution while co-administration with this drug. (Agrawal et.al 2020).Some studies shows that there is binding of the F-RTP to the active site of the SARS-CoV and MERs-CoV in the presence of the agent of protein as the F-RTP bound to RNA terminal in the presence of the magnesium ions, nucleotide phosphate and RdRp proteins (NSP-12, NSP-7and NSP-8). This drug bind to the protein of the SARS-CoV-2 to the NPS-15 protein which is endonuclease of the SARS-CoV-2 plays an important role in the replication of the virus RNA. This distinct mechanism underlying of favipiravir-mediated interaction with corona virus RdRp.

Mechanism of the Favipiravir:

When the covid-19 virus enters into the host cell, tissue with the molecule undergoes phosphor-ribosylation to Favipiravir-RTP which is the active form of this drug which shows the antiviral drug effect on the SARS-CoV-2 virus. This molecule act as the substrate for the RNA-dependent RNA-polymerase (RdRp) enzyme which not taken the purine nucleotide in the SARS corona virus thus inhibiting activity of this enzyme lead to the termination of the viral protein synthesis. As this enzyme get incorporated with the viral RNA strand which inhibit the preventing the further extension and stop the replication of the SARS-CoV-2. The mechanism of action along with preservation of the catalytic domain of the RdRp enzyme across the various virus show broad spectrum of this drug (Sada *et al.,* 2020).

Application of Flavipiravir:

Purine nucleotide from the Favipirin drug inhibits the enzyme activity of which lead to the termination of the viral protein synthesis.

Side effects of the Flavipiravir:

Some Japanese studies observed that, there are acute reaction seen in the individual who are treated with these drugs they show relatively minor and include the Hyperuricemia and diarrhea in the 5% of the patient and there is reduction in Neutrophile count in 2% percent of the patients.

Side effects	>1%	0.5-<1%	0.5%
Hyper-sensitivity		Rash	Eczeme. pruitus
Gastro-intestinal	Diarrhea	Nausea, vomiting, adominal pain	Adominal discomfort dudodenal ulcer, heematochezia gastritis
Hopatic	AST(GOT) increased ALT (GPT) increased γ -GTP increased		Blood ALP increased blood hillirubin increased
Hemotologic	Neutrophil count decreased, white blood cell decreased	Glucose in the urine present	White blood cell count increased reticulocytes count decreases monocytes increased.
Metabolic disorder	Blood uric acid increasedBlood triglycerides increased		Blood potassium decreased.
Respiratory			Asthma, Oropharyngeal, pain,rhinitis, Nasopharygitis
Other			CPk increase blood urine present tonsil poly pigmentation, Dysgeusis, bruise.

Convalescent plasma therapy treatment:

Convalescent plasma therapy has been classic adaptive immuno-therapy applied to the prevention and treatment of Covid-19 and various kinds of diseases. As the pathogen enter into the body of the individual there are two lines of defense i.e. innate immunity and the acquired immunity. The innate immunity contains number of the soluble cell-based antimicrobial factors, and they get triggered as soon as the infection takes place in the individual.Adaptive immunity consists of the pathogen-specific antibody and T-cells and develops later and contribute for fighting with pathogen and to eliminate the activity of the pathogen. In addition to the adaptive and innate immune system, some of the immuno-modulators, immuno-suppressants drug can also stimulate immune system and helps to remove the pathogen or suppress the activity of the pathogen (Rizk *et al.,* 2020) lives. So one dose of the 200 ml of the convalescent plasma can help an individual to fight for the covid-19 virus and eliminate it or suppress its effect. While clinical symptoms significantly improved with the increase in the oxygen level of an individual within approximately 3 days as we know that corona virus has one of the rapid viral replication system and ability for massive inflammatory cell infiltration and elevated pro-inflammatory cytokines or even cytokines storm in the alveoli of lungs results in the acute pulmonary injury and acute respiratory distress.

As the recent, studies found that in covid-19 that lymphocytes count in the peripheral blood were reduced and the level of the cytotoxin in the plasma requires the ICU support IL-6, IL-10. Recovered Covid-19 patients contain the large number of the humoral immunity against the virusand large neutralizing antibody which is highly capable for neutralizing the SARS-CoV-2 and remove the pathogen from the blood and from the pulmonary tissue too. There is decrease in the antibody of the SARS-CoV-2 decrease gradually after the 4 months after disease process. If the antibody tiler decreases rapidly within the 3monthshence, study suggest that the humoral immune response from the plasma of the recently recovered patient more is effective. The recently recovered Covid-19 patient has neutralizing antibody tiler above the 1:640 and hence they are the suitable for the plasma donation for saving the other patient life by plasma therapy (Duan *et al.,* 2020).

Conclusion:

From the above detail study of various therapeutic drugs and therapy it is clear that all the drugs are highly capable for the controlling the covid-19 virus However, when the covid-19 emerged in the world, these drugs have played a critical role of saving lots of lives.So,these drugs have proved as the life-saving drugs to some extent.

References:

1. Damir Hurenovic (2019). Brief History of pandemic, Psychiatry of pandemic volume 1: 1-12.

2. Harapan Harapan, Naoya Itoh, Amanda Yufika,Wira Winardi, Synat Keam, HayphengTe, Dewi Megawati, Zinatul Hayati, Abram L. Wagner and Mudatsir Mudatsir, (2020). Coronavirus disease 2019 (COVID-19): A literature review, Journal of Infection and Public Health, Volume 13 (5): 667-673.

3. https://covid19.who.int/)

4. Anshumali Mittal, Kavyashree Manjunath, Rajesh Kumar Ranjan, Sandeep Kaushik, Sujeet Kumar, and Vikash Verma(2020).COVID-19 pandemic: Insights into structure, function, and hACE2 receptor recognition by SARS-CoV-2, PLOS Pathogens,Vol. 16 (8): 2-6.https://doi.org/10.1371/journal.ppat.1008762

5. S. Saqrane, M.A. El Mhammedi, S. Lahrich, F. Laghrib, Y. El Bouabi, A. Farahi and M. Bakasse, (2020). Recent knowledge in favor of remdesivir (GS-5734) as a therapeutic option for the COVID-19 infection Journal of Infection and Public Health,Vol. 14 (1): 1-25.

5. https://www.cdc.gov/coronavirus/2019-ncov/symptoms-testing/symptoms.html

6. Marco Cascella, Michael Rajnik, Abdul Aleem, Scott C. Dulebohn and Raffaela DiNapoli (2021). Features, Evaluation, and Treatment of Coronavirus (COVID-19) National Library of Medicine, National Institutes of Health. StatPearls,1-17.

7.Markus Hoffmann, Hannah Kleine Weber, Simon Schroeder, Nadine Krüger, Tanja Herrler, Sandra Erichsen, Tobias S.Schiergens, Georg Herrler, Nai-Huei Wu, Andreas Nitsche, Marcel A.Müller, Christian Drosten and Stefan Pöhlmann, (2020). SARS-CoV-2 Cell Entry Depends on ACE2 and TMPRSS2 and Is Blocked by a Clinically Proven Protease Inhibitor, Cell, Vol. 181(2): 271-280.

8. Chen Y, Liu Q and Guo D. (2020). Emerging coronaviruses: genome structure, replication, and pathogenesis. J Med Virol. Vol. 92(4):418-423. doi:10.1002/jmv. 25681.

9. Fehr A. R. and Perlman S. (2015). Coronaviruses: an overview of their replication and pathogenesis. Methods Mol Biol. 1282:1-23. doi:10.1007/978-1-4939- 2438-7_1

10. Fung TS and Liu DX (2014). Coronavirus infection, ER stress, apoptosis and innate immunity. Front Microbiol. Vol. 5(296): 1-13. doi:10.3389/fmicb.2014. 00296.

11. James M. Sanders, Marguerite L. Monogue, Tomasz Z. Jodlowski, James B. and Cutrell, MD (2020). Pharmacologic Treatments for Coronavirus Disease 2019 (COVID-19) A Review, JAMA, 323(18):1824-1836.

12. Richard T. Eastman, Jacob S. Roth, Kyle R. Brimacombe, Anton Simeonov, Min Shen, Samarjit Patnaikand Matthew D. Hall (2020). Remdesivir: A Review of Its Discovery and Development Leading to Emergency Use Authorization for Treatment of COVID-19, ACS Central Science, Vol 6 (5): 672-683.

13. Calvin J. Gordon, X Egor P. Tchesnokov, X Emma Woolner, X Jason K. Perry, Joy Y. Feng, Danielle P. Porter, and Matthias Gotte, (2020). Remdesivir is a direct-acting antiviral that

inhibits RNAdependent RNA polymerase from severe acute respiratory syndrome coronavirus 2 with high potency journal of Biological chemistry, 295 (20):6785-6797.

14. Samia Arshad, Paul Kilgore, Zohra S. Chaudhry, Gordon Jacobsen, Dee Dee Wang, Kylie HuitSing, Indira Brar, George J. Alangaden, Mayur S. Ramesh, John E. McKinnon, William O'Neill and Marcus Zervos and Henry Ford, (2020).Treatment with hydroxychloroquine, azithromycin, and combination in patients hospitalized with COVID-19,International Journal of Infectious Diseases, 97: 396-403.

15. Sairaj Satarker, Tejas Ahuja, Madhuparna Banerjee, Vignesh Balaji E, Shagun Dogra, Tushar Agarwal and Madhavan Nampoothiri, (2020). Hydroxychloroquine in COVID-19: Potential Mechanism of Action Against SARS-CoV-2, Current pharmacology reports,Vol. 6: 203-211.

16. Lo'ai Alanagreh, Foad Alzoughool and Manar Atoum, (2020). Risk of using Hydro-chloroquine as a treatment of Covid-19, International Journal of Risk and Safety in Medicine, Vol. 31(3): 111-116.

17. Umang Agrawal, Reyma Raju, and Zarir F. Udwadia (2020) Favipiravir: A new and emerging antiviral option in COVID-19, Medical Journal of Armed Forces India, Vol. 76: 370-376.

18. Mitsuru Sada, Takeshi Saraya, Haruyuki Ishii, Kaori Okayama, Yuriko Hayashi, Takeshi Tsugawa, Atsuyoshi Nishina, Koichi Murakami, Makoto Kuroda, Akihide Ryo and Hirokazu Kimura(2020). Detailed Molecular Interactions of Favipiravir with SARS-CoV-2, SARS-CoV, MERS-CoV, and Influenza Virus Polymerases In Silico,Microorganisms, Vol. 8 (1610): 1-9.https://doi.org/10.3390/microorganisms8101610.

19. John G. Rikz, Kamyar kalantar Zadeh, Mandeep R. Mehra, Carl J. Lavie, Youssef Rizk , and Donald N Forthal (2020), Pharmaco-Immunomodulatory Therapy In covid -19,Drugs, 80 (13): 1267-1292.

20.Kai Duan, Bende Liu, Cesheng Li,, Huajun Zhang, Ting Yu, Jieming Qu, Min Zhou, Li Chen, Shengli Meng, Yong Hu, Cheng Peng, Mingchao Yuan, Jinyan Huang, Zejun Wang, Jianhong Yu, Xiaoxiao Gao, Dan Wang, Xiaoqi Yu, Li Li, Jiayou Zhang, Xiao Wu, Bei Li, Yanping Xu, Wei Chen, Yan Peng, Yeqin Hu, Lianzhen Lin, Xuefei Liu, Shihe Huang, Zhijun Zhou, Lianghao Zhang, Yue Wang, Zhi Zhang, Kun Deng, Zhiwu Xia, Qin Gong, Wei Zhang, Xiaobei Zheng, Ying Liu, Huichuan Yang, Dongbo Zhou, Ding Yu, Jifeng Hou, Zhengli Shi , Saijuan Chen, Zhu Chen,Xinxin Zhang and Xiaoming Yang, (2020). The feasibility of convalescent plasma therapy in severe COVID19 patients: a pilot study: 1-22.